

Elia Jean Botros

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EDUCATION

The Pennsylvania State University, University Park

Expected Graduation: December 2026

Bachelor of Science in Industrial Engineering

GPA: 3.3

Minor: Engineering Leadership Development

WORK EXPERIENCE

Kew Gardens Animal Hospital, Queens, New York

Summers 2021-2025

Animal Hospital Assistant

- Prepared animals and equipment for examinations and surgeries, supporting efficient clinical workflows.
- Maintained cleanliness and organization of equipment and treatment areas through consistent sterilization practices.
- Assisted with front-desk operations, patient intake, and scheduling to support smooth clinic operations.

PROJECT EXPERIENCE

Engineering Project Management Experience

Sept. 2025 – Dec. 2025

Project Lead and Team Member

- Developed detailed project schedules using Microsoft Project, including task dependencies, milestones, and critical path analysis.
- Created Work Breakdown Structures, assigned resources, and established baseline schedules to track project progress.
- Conducted risk analysis, change control activities, and formal project closeout with lessons learned documentation.

ACRP University Design Competition

Jan. 2025 – May 2025

Team Lead & System Designer

- Designed an AI-driven route optimization system to reduce airport ground equipment emissions using MATLAB and SUMO simulation software.
- Collaborated with a 4-member team to develop data-driven routing algorithms and evaluate environmental impact across multiple traffic scenarios.
- Conducted stakeholder interviews to refine solution feasibility
- Integrated simulation results into performance metrics.

LEADERSHIP & INVOLVEMENT

Mentor & Tutor, FIRST Robotics Competition Team 2638

Jan. 2024 – May 2024

- Provided assistance in mechanical, electrical, and systems-level robotics concepts.
- Assisted students in understanding design tradeoffs and competition requirements.

Broadcast Lead, Penn State Intramural Sports

Jan. 2023 – Oct. 2023

- Led live broadcast operations, coordinating technical setup, commentary, and production flow.
- Managed real-time problem solving and team communication during live events.

SOFTWARE SKILLS

MATLAB, Python, SQL, Microsoft Excel, SolidWorks, AutoCAD, Microsoft Project, Minitab, LINDO, Figma